

Grade 2 Mathematics

Մաթեմատիկա — 2-րդ դասարան

COURSE CODE	MTH-G2
PROGRAM	Elementary Mathematics · Տարրական մաթեմատիկա
LEVEL	Grade 2 · typical age 7-8
DURATION	60-72 instructional hours (approx. 60 lessons)
PREREQUISITE	Grade 1 Mathematics or equivalent: fluency within 20, place value to 120.
LEADS TO	Grade 3 Mathematics (MTH-G3)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 2 extends place value to 1,000 and turns the strategies of Grade 1 into reliable multi-digit procedures. Students add and subtract within 100 fluently and within 1,000 with models, learn to measure with standard units and to estimate, work with equal groups and arrays as the seed of multiplication, and read and construct simple data displays. The year ends with a student who can compute accurately and, just as importantly, can say why the computation works.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

The central risk of Grade 2 is that a child learns the standard algorithm as a ritual — a sequence of marks with no meaning. Վարժարան teachers introduce every multi-digit procedure through place-value materials and expanded notation first, and require the student to explain a regrouping in terms of trading ten ones for one ten before the compact written form is permitted. Understanding is not a supplement to procedure here; it is the thing that makes procedure survive to Grade 4.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Adds and subtracts within 20 without finger counting	If absent, revisit Grade 1 Unit 3 before beginning; multi-digit work will otherwise collapse.
Explains that the 4 in 47 means four tens	Place-value understanding, not just numeral reading, is the prerequisite for regrouping.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves a one-step word problem within 20 and writes an equation for it	Indicates the child can move between situation and symbol.
Counts by 10s from any number	Prerequisite for adding tens and for skip counting toward multiplication.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Understand place value to 1,000, including hundreds, tens, and ones, and count within 1,000.
- 2 Fluently add and subtract within 100 using strategies based on place value and properties of operations.
- 3 Add and subtract within 1,000 using concrete models and written strategies, and explain the reasoning.
- 4 Know all sums of two one-digit numbers from memory.
- 5 Solve one- and two-step word problems within 100 involving all four problem types.
- 6 Measure and estimate length in inches, feet, centimeters, and meters, and choose an appropriate tool.
- 7 Work with equal groups, arrays, and repeated addition as a foundation for multiplication.
- 8 Tell time to the nearest five minutes and solve problems with dollar bills and coins.
- 9 Draw a picture graph and a bar graph and answer comparison questions from them.

COURSE UNITS

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Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 7-8 LESSONS

Place Value to 1,000

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TOPICS COVERED

- Bundling ten tens into one hundred
- Three-digit numbers as hundreds, tens, and ones
- Standard, expanded, and word form
- Counting within 1,000 by 1s, 5s, 10s, and 100s
- Skip counting from any starting number
- Comparing three-digit numbers with $>$, $<$, $=$
- 10 more/less and 100 more/less, mentally

LEARNING OBJECTIVES

- 1 Explain 100 as a bundle of ten tens.
- 2 Write any number to 1,000 in standard, expanded, and word form.
- 3 Compare three-digit numbers by reasoning from the largest place downward.
- 4 Mentally add or subtract 10 or 100 to any three-digit number.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Converts fluently among the three forms in 9 of 10 trials.
- Compares three-digit numbers correctly in 9 of 10 trials with place-value justification.
- Answers '100 less than 428' in under 4 seconds.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child writes 'three hundred forty-two' as 30042.	Concatenation error — the child is writing what they hear. Build the number with base-ten blocks and write expanded form as an intermediate step.
Zero as a placeholder is dropped: 405 written as 45.	Explicitly practice numbers with internal zeros; use place-value charts where the empty column must be filled.

TEACHING NOTE

Everything in Grade 2 rests on this unit. Base-ten blocks — physical or on the shared whiteboard — should be visible in every lesson of the unit.

UNIT 2 • 9-10 LESSONS

Addition and Subtraction Within 100

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TOPICS COVERED

- Adding and subtracting tens
- Adding two-digit numbers without regrouping
- Regrouping ones into a ten: the meaning before the notation
- The standard algorithm for addition
- Regrouping a ten into ones for subtraction
- The standard algorithm for subtraction
- Mental strategies: compensation, adding up, near-doubles
- Checking subtraction with addition
- Adding four two-digit numbers

LEARNING OBJECTIVES

- 1 Add and subtract within 100 fluently, choosing between mental and written methods.
- 2 Explain a regrouping as a trade of ten ones for one ten, using blocks or drawings.
- 3 Use compensation (e.g., $48 + 27$ as $50 + 25$) as a mental strategy.
- 4 Verify a subtraction result by adding.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes 20 two-digit sums and differences with 90% accuracy in 5 minutes.
- Explains one regrouping step aloud in place-value language.
- Uses at least one mental strategy correctly without prompting.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child subtracts the smaller digit from the larger regardless of position ($52 - 27 = 35$).	The most common error in Grade 2. Model with blocks: you cannot take 7 from 2, so you must unbundle a ten. Never let the child proceed on the written form until the block model is correct.
The regrouped 1 is written but never added.	Slow down and name it: 'that one is one ten, and it belongs in the tens column.' Use color to mark it.

TEACHING NOTE

Fluency within 100 is the expected exit skill. Fluent means accurate, reasonably quick, and flexible — a student who only knows the algorithm is not yet fluent.

UNIT 3 • 7-8 LESSONS

Addition and Subtraction Within 1,000

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TOPICS COVERED

- Adding three-digit numbers with base-ten models
- Regrouping in two places
- Subtracting across a zero
- Written strategies and the standard algorithm extended
- Estimating sums and differences by rounding to the nearest hundred
- Judging whether an answer is reasonable

LEARNING OBJECTIVES

- 1 Add and subtract within 1,000 using models and written strategies, explaining each regrouping.
- 2 Subtract across a zero correctly (e.g., $400 - 137$).
- 3 Estimate a sum before computing and use the estimate to check the result.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves three-digit problems with regrouping in two places at 85% accuracy.
- Solves 4 of 5 subtract-across-zero problems correctly.
- Produces a reasonable estimate before computing in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Subtracting across a zero produces errors like $400 - 137 = 373$.	Model unbundling twice with blocks: one hundred becomes ten tens, one ten becomes ten ones. Rushing this creates a gap that persists to Grade 5.
The child estimates <i>after</i> computing, to match the answer.	Require the estimate to be written down and circled before any computation begins.

TEACHING NOTE

Estimation is not decoration. A student who estimates first catches their own place-value errors, which is the whole point.

UNIT 4 • 7-8 LESSONS

Measurement, Length, and the Number Line

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TOPICS COVERED

- Measuring with rulers, yardsticks, meter sticks, and tapes
- Inches and feet; centimeters and meters
- Choosing an appropriate unit and tool
- Estimating lengths before measuring
- Comparing lengths and finding the difference
- Adding and subtracting lengths within 100
- Representing whole numbers as points on a number line
- Solving length problems on a number line

LEARNING OBJECTIVES

- 1 Measure the same object in two different units and explain why the numbers differ.
- 2 Estimate a length in a given unit before measuring, then check.
- 3 Use a number line diagram to represent and solve an addition or subtraction problem.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Measures to the nearest whole unit accurately in 4 of 5 trials.
- Explains why measuring in centimeters gives a larger number than inches.
- Correctly places and uses jumps on a number line for a two-step problem.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Measurement begins at the end of the ruler rather than at the zero mark.	Deliberately hand the child a broken ruler starting at 3; ask how to measure with it. This forces attention to the zero point.
A larger unit is assumed to give a larger number.	Measure a desk in inches and in feet side by side. The inverse relationship must be experienced, not told.

TEACHING NOTE

The number line introduced here becomes the backbone of fractions in Grade 3 and of integers in Grade 6. Treat it as a major idea, not a diagram.

UNIT 5 · 6-7 LESSONS

Foundations of Multiplication

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TOPICS COVERED

- Equal groups and the language of 'groups of'
- Repeated addition
- Building and describing rectangular arrays
- Rows and columns; finding the total two ways
- Skip counting by 2s, 5s, and 10s as repeated addition
- Even and odd numbers via pairing
- Writing an equation for an array as a sum of equal addends

LEARNING OBJECTIVES

- 1 Represent an equal-groups situation with objects, an array, and a repeated-addition equation.
- 2 Find the total in an array by counting rows and by counting columns, and explain why both work.
- 3 Determine whether a number is even or odd by pairing, and justify.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Builds an array for a given equal-groups story in 4 of 5 trials.
- Writes a correct repeated-addition equation for an array.
- Classifies numbers to 20 as even or odd with justification.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child treats a 3×4 array and a 4×3 array as unrelated.	Physically rotate the array. Seeing the same object from two orientations is the concrete root of commutativity.
'Even' is defined as 'ends in 0, 2, 4, 6, 8' with no meaning attached.	Define even as 'can be split into two equal groups with none left over' and let the digit pattern be a discovered consequence.

TEACHING NOTE

No multiplication symbol is required this year, but the array model introduced here is the same model used for multi-digit multiplication in Grade 4 and for area in Grade 3. Build it well.

UNIT 6 • 7-8 LESSONS

Time, Money, Data, and Shapes

Ժամանակ, փող, տվյալներ և պատկերներ

TOPICS COVERED

- Telling time to the nearest five minutes; a.m. and p.m.
- Elapsed time in whole and half hours
- Counting mixed collections of coins and bills
- Solving money word problems with \$ and ¢ symbols
- Generating measurement data and making a line plot
- Picture graphs and bar graphs with single-unit scales
- Answering put-together, take-apart, and compare questions from a graph
- Recognizing and drawing shapes by number of angles and faces
- Partitioning rectangles into equal squares

LEARNING OBJECTIVES

- 1 Tell and write time to the nearest five minutes on analog and digital clocks.
- 2 Count a mixed collection of coins and bills and write the amount correctly.
- 3 Draw a bar graph from a data set and answer a comparison question about it.
- 4 Partition a rectangle into rows and columns of same-size squares and count them.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Reads five-minute times correctly in 9 of 10 trials.
- Counts a mixed coin collection under \$2.00 accurately in 4 of 5 trials.
- Constructs a correct bar graph with labeled axes from raw data.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Coins are counted by quantity rather than by value.	Sort by value from largest to smallest before counting, and count on: 25, 35, 45, 50.
Bar graph bars are drawn without a consistent scale.	Require axis labels and a stated scale on every graph, every time, from the first example.

TEACHING NOTE

Partitioning rectangles into same-size squares is quietly the beginning of area. Connect it aloud to the arrays of Unit 5.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Grade 1 exit skills plus early Grade 2 content; produces the initial academic profile.
Unit Check	End of each unit · 15 min	That unit's mastery criteria; oral explanation is scored alongside accuracy.
Fluency Probe	Every 3rd lesson · 3 min	Sums within 20 from memory, then two-digit computation.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3, weighted toward place value and regrouping.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Grade 3 placement and pacing.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 45–50 min	Full course in approximately 40–45 weeks.
Accelerated	2 lessons/week	Full course in approximately 22–25 weeks.
Intensive / Catch-Up	2–3 lessons/week	16–20 weeks; typically Units 1–3 first, then the remainder.
Summer Program	3 lessons/week, 6 weeks	Regrouping intensive (Units 2–3) or full-year preview.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Fifteen minutes per lesson, six to eight problems. Every assignment contains at least two problems from a prior unit — spaced retrieval is what keeps Grade 1 fluency from decaying while Grade 2 content is being learned. Word problems are always included, never postponed to the end of the year. Parents record the time spent and any problem that caused frustration; the teacher reviews this before the next lesson.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Reads, writes, and compares any number to 1,000 in three forms.
- ◆ Knows all sums of two one-digit numbers from memory.
- ◆ Adds and subtracts fluently within 100.
- ◆ Explains regrouping in place-value language, not as a rule.
- ◆ Adds and subtracts within 1,000, including across a zero.
- ◆ Measures and estimates length in both customary and metric units.
- ◆ Represents equal groups with arrays and repeated addition.
- ◆ Tells time to five minutes and counts mixed money.
- ◆ Completed Grade 2 Mathematics — ready for Grade 3.

MATERIALS AND RESOURCES

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- Base-ten blocks (hundreds flats, tens rods, ones cubes) — physical set at home and digital on the shared board.
- Rulers in inches and centimeters; a measuring tape.
- US coins and bills (play money) for the money unit.
- Վարժարան Grade 2 problem bank — approximately 800 items tagged by unit, type, and difficulty.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.